

Introduction

This project focuses on analyzing retail transactional data to uncover profit trends, regional performance, and seasonal sales behavior using SQL, Python, and Tableau.

Abstract

The project performs data cleaning, SQL analysis, exploratory data analysis (EDA), and dashboard visualization to derive meaningful business insights and support data-driven decision-making.

Tools Used

- SQLite
- Python
- Pandas
- Seaborn
- Tableau Public
- Jupyter Notebook

Page 2

Steps Involved

1. Imported retail dataset into SQLite
2. Performed SQL analysis queries
3. Conducted EDA using Python
4. Built correlation heatmaps and trend charts
5. Created interactive Tableau dashboard
6. Generated business insights

Key Insights

- Technology category had highest profitability
- West region generated maximum sales
- Sales increased during year-end months
- Discounts negatively affected profitability

Conclusion

The project successfully demonstrated how SQL, Python, and Tableau can be used together to analyze retail business performance and generate actionable insights.

